

# Advanced Algorithms

## First Programming Project 2026/27

### DI – NOVA FCT

The first programming project is completed in teams of **two** students and consists of a theoretical and experimental comparative study of alternative approximation algorithms for a **single** optimization problem. Each group must choose the optimization problem to study and at least two approximation algorithms for the chosen problem.

The report should contain the following topics:

- **Problem:** Define the problem and indicate the complexity class of the corresponding decision problem. The proof is not required.
- **Algorithms:** For each (chosen) approximation algorithm, describe the algorithm, analyse its time and space complexities, and specify its approximation ratio. The proof of the approximation ratio is not required, but a reference for it must be provided.
- **Tests:** Characterise the instances used in the experiments, specifying whether they come from available benchmarks or have been created by you.
- **Test Results:** Present and analyse the experimental results, comparing the implemented approximation algorithms. The analysis should take into account execution times.
- **AI Tools:** Specify which websites were consulted and which AI tools were used, for what purposes, and how the information obtained was used to create the source code, the tests, and the report.

Send the report (as a PDF file), the source code, and the input files containing the instances used in the experiments to Margarida Mamede ([mm@fct.unl.pt](mailto:mm@fct.unl.pt)). The **deadline** is **October 25th** (Sunday).

Here are some possible optimization problems for which approximation algorithms are available. You may (but do not have to) choose one of them.

- |                              |                                |
|------------------------------|--------------------------------|
| • Min. Bin Packing           | • Min. Load Balancing          |
| • Min. Dominating Set        | • Min. (Weighted) Set Cover    |
| • Min. k-Clustering          | • Min. (Metric) Steiner Trees  |
| • Min. (Weighted) Edge Cover | • Max. Subset Sum              |
| • Max. Independent Set       | • Metric Travelling Salesman   |
| • Knapsack                   | • Min. (Weighted) Vertex Cover |